Prathit Sharma

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Mechanical Engineering undergraduate working across mechanical design, computational modelling and strategy. Hands-on exposure to aerospace hardware manufacturing, rocket-propulsion research and hydropower O&M, with parametric MATLAB/Python models behind every design decision and a consulting track record through 180 Degrees Consulting and national case competitions.

EDUCATION

Degree/Certificate	Institute/Board	Score	Year
B.Tech. – Mechanical Eng.	Indian Institute of Petroleum and Energy	CGPA: 7.3/10	2024–Present

SKILLS

- **Engineering Domains:** Fluid Mechanics, Thermodynamics, Turbomachinery, Systems Engineering, Trade Studies, Euler Buckling Analysis, Pneumatic System Sizing, Thermochemical Analysis.
- **Programming & Computation:** MATLAB (Parametric Sweeps), Python (pandas, SciPy), C.
- **Design & Tools:** SolidWorks, Fusion 360, AutoCAD, NASA CEA, MS Excel, LaTeX, Git.
- **Quality & Business:** Six Sigma (DMAIC), QMS & Technical Documentation, Financial Modelling (NPV, payback), Monte Carlo Risk Analysis.

INTERNSHIPS

SEC Industries Private Limited **May 2026 – June 2026**
Engineering Intern *Hyderabad, India*

– Rotated through Design, QC, Fabrication, Machine Shop and Planning; worked with the QMS, technical documentation and traceability records.
– Conducted preliminary mechanical design, trade studies and structural optimization for a high-speed pneumatic UAV catapult launcher.

Bharat Space Education Research Centre (BSERC) **June 2026 – July 2026**
Def-Space Tech Intern

– Researched pintle-type liquid bipropellant (LOX/LCH4) injector designs, deriving the foundational numerical parameters for atomization and combustion stability.
– Studied autonomous UAV architectures and defense-tech systems, applying propulsion and safety-critical design fundamentals.

NHPC Ltd. – Salal Power Station **Dec 2025**
Operations & Maintenance Intern (Capital Maintenance Unit) *Reasi, J&K*

– Supported Capital Maintenance of Unit 6 (115 MW) at a high-silt river station; inspected hydraulic path integrity and documented silt-induced runner erosion, evaluating HVOF coatings for component longevity.

PROJECTS

Movable Pintle Injector Design for LOX/LCH4 Liquid Rocket Engine **July 2026 – Present**
Independent Research – paper accepted at NCCS 2026 link

– **Novel Manifold Architecture:** Conceptualized a throttleable propellant manifold applying flow-distribution principles from Francis turbine spiral casings to reduce circumferential swirl and uneven spray.
– **Numerical & Thermochemical Analysis:** Authored a MATLAB sizing script using NASA CEA to set a baseline design point (O/F = 3.2, 750 N thrust, 20 bar).

High-Speed Pneumatic UAV Catapult Preliminary Design **May 2026 – June 2026**
SEC Industries Private Limited (Internship Project) link

– **Parametric Sizing:** Built a 396-point MATLAB sweep (11 launch angles × 36 UAV masses); the governing case (20 kg at 15°) drove a 125 mm bore with a buckling factor of safety of 5.5.
– **Component Specification:** Sized ISO 6431 cylinders and EN 286-1 accumulators and verified all 396 conditions against the selected hardware.

Break The Case '26 – Six Sigma Invoice-Error Reduction **Oct 2026**
Team Case Competition, BITS Hyderabad (Round 2) link

– **DMAIC Analysis:** Diagnosed supplier invoice errors (10,648 DPMO) and built a Python pipeline that searched 25,599 solution combinations and ran 20,000 Monte Carlo scenarios.
– **Recommendation:** Phased hybrid plan projecting a 90% error cut, Rs. 162 lakh 3-year NPV and 9-month payback in the base case.

LEADERSHIP

- **Founding President, 180 Degrees Consulting IIPE** – founded the chapter, ran its first recruitment drive and lead pro-bono NGO engagements.
- **Social Welfare Secretary, IIPE & NSS Student In-charge** (2025–26); **Co-founder, Sattva Quizzing Club, IIPE**. **Award:** Socionnovate National Finalist.